

Week 5

1. Find the vector equation for the line through $\overset{P}{(-7, 6, 0)}$ and $\overset{Q}{(-1, 1, 4)}$. Recall that to find the vector equation for a line through two points (a, b, c) and (d, e, f) , you find a direction vector of the form $[d - a \ e - b \ f - c]$ and then the equation of the line is $[x \ y \ z] = [a \ b \ c] + t[d - a \ e - b \ f - c]$ where $t \in \mathbb{R}$. Then, find the parametric equations for this line.

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix} - \begin{bmatrix} -7 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -7 \\ 6 \\ 0 \end{bmatrix} + t \begin{bmatrix} 6 \\ -5 \\ 4 \end{bmatrix}, t \in \mathbb{R} \leftarrow \text{vector form}$$

$$\begin{cases} x = -7 + 6t \\ y = 6 - 5t \\ z = 4t \end{cases}, t \in \mathbb{R} \leftarrow \text{Parametric form}$$

2. Find the vector equation for the line through the two points $\overset{P}{(-5, 5, 1)}$, $\overset{Q}{(2, 2, 4)}$. Then, find the parametric equations.

$$\overrightarrow{PQ} = \begin{bmatrix} 7 \\ -3 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -5 \\ 5 \\ 1 \end{bmatrix} + t \begin{bmatrix} 7 \\ -3 \\ 3 \end{bmatrix}, t \in \mathbb{R}$$

$$\begin{cases} x = -5 + 7t \\ y = 5 - 3t \\ z = 1 + 3t \end{cases}, t \in \mathbb{R}$$

3. The equation of a line in two dimensions is written as $y = x - 5$. Find parametric equations for this line.

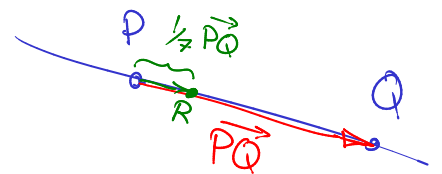
$$y = x - 5 = t \quad t \in \mathbb{R} \rightsquigarrow \begin{cases} y = t \\ x - 5 = t \end{cases} \rightsquigarrow \begin{cases} y = t \\ x = 5 + t \end{cases}, t \in \mathbb{R}$$

4. Find the point on the line segment from $P = (-4, 7, 5)$ to $Q = (2, -2, -3)$ which is $\frac{1}{7}$ of the way from P to Q .

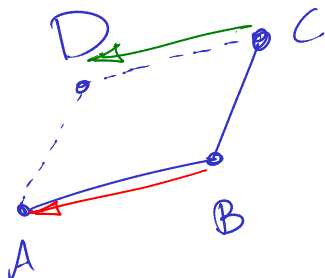
$$\vec{OR} = \vec{OP} + \frac{1}{7} \vec{PQ}$$

$$\vec{PQ} = \begin{bmatrix} 6 \\ -9 \\ -8 \end{bmatrix}$$

$$\vec{OR} = \begin{bmatrix} -4 \\ 7 \\ 5 \end{bmatrix} + \frac{1}{7} \begin{bmatrix} 6 \\ -9 \\ -8 \end{bmatrix} = \begin{bmatrix} -4 + \frac{6}{7} \\ 7 - \frac{9}{7} \\ 5 - \frac{8}{7} \end{bmatrix} \rightarrow R = \left(-\frac{22}{7}, \frac{40}{7}, \frac{27}{7} \right)$$



5. Given $A = (2, 1, 0)$, $B = (-3, 1, 3)$, and $C = (1, 3, -1)$, find D so that $ABCD$ is a parallelogram.



Find the vector $\vec{BA}$.

Then from C move along $\vec{BA}$.

That is,

$$\vec{OD} = \vec{OC} + \vec{BA}$$

$$\vec{OD} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 2+3 \\ 1-1 \\ 0-3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ -4 \end{bmatrix}$$

then D is $(6, 3, -4)$.